EXPERIMENT 4

AIM: To develop a blockchain-powered web application using Solidity programming language on Remix IDE and embedding Metamask Wallet

THEORY:

Smart Contract:
A smart contract is a program stored on the blockchain that automatically executes predefined operations when its functions are called. In this experiment, the smart contract is written using Solidity.

Solidity:
Solidity is a programming language used to develop smart contracts for Ethereum-compatible blockchain networks.

Remix IDE:
Remix is a web-based IDE used to write, compile, deploy, and interact with Solidity smart contracts.

MetaMask:
MetaMask is a cryptocurrency wallet that allows users to manage blockchain accounts and connect decentralized applications to blockchain networks. It is used to approve and sign transactions in this experiment.

Testnet:
A testnet is a blockchain network used for development and testing. It uses test cryptocurrency, such as Sepolia ETH, which does not have real monetary value.

Transaction:
A blockchain transaction is an operation submitted to the network. Deployment and state-changing smart contract functions require a transaction and are confirmed through MetaMask.

TASKS PERFORMED

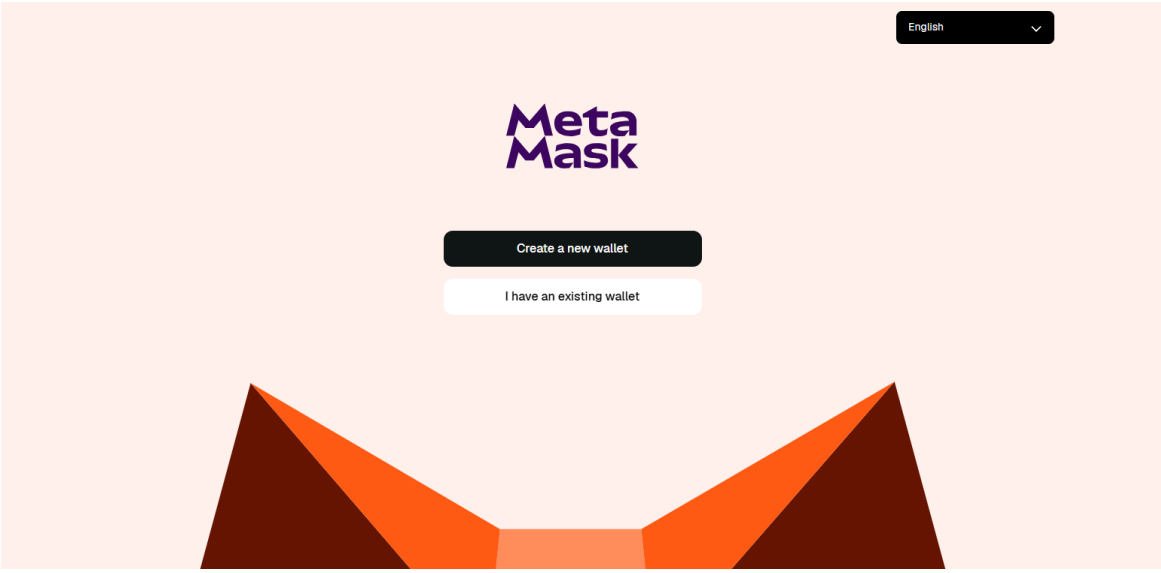
1. Installed and configured the **MetaMask Wallet**.
2. Created/imported a MetaMask account and connected it to the **Sepolia/RSK Testnet**.
3. Obtained test cryptocurrency from the testnet faucet.
4. Created a simple **Solidity smart contract** in Remix IDE.
5. Compiled and deployed the smart contract using MetaMask.
6. Interacted with the deployed smart contract and performed transactions.
7. Verified the transaction details using the blockchain explorer.

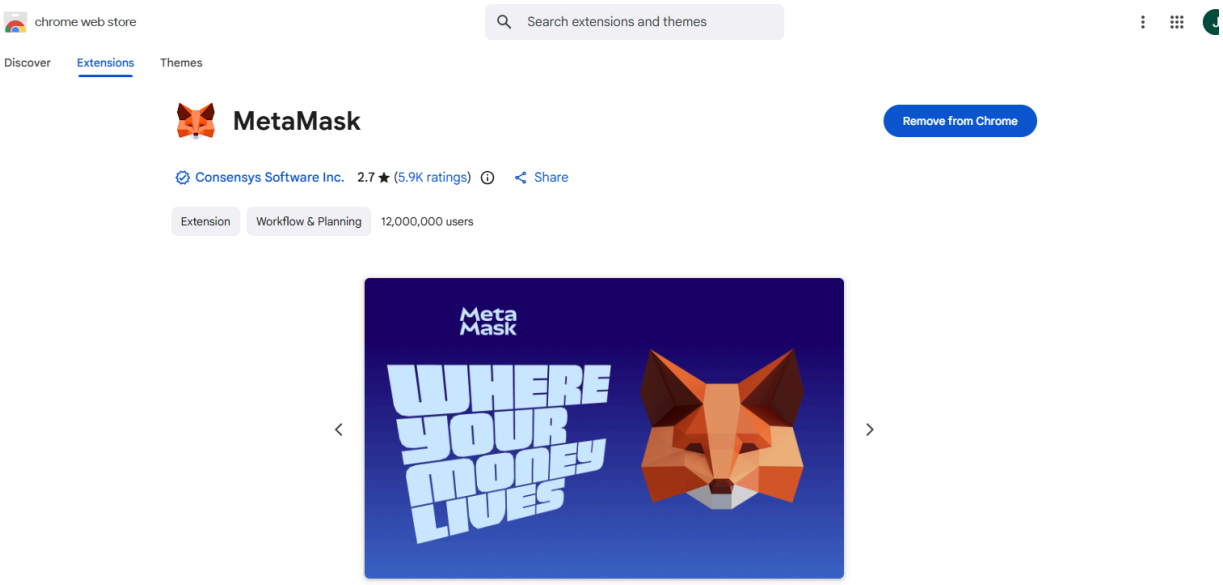
PROCEDURE:

Step - 1 : Set Up MetaMask:

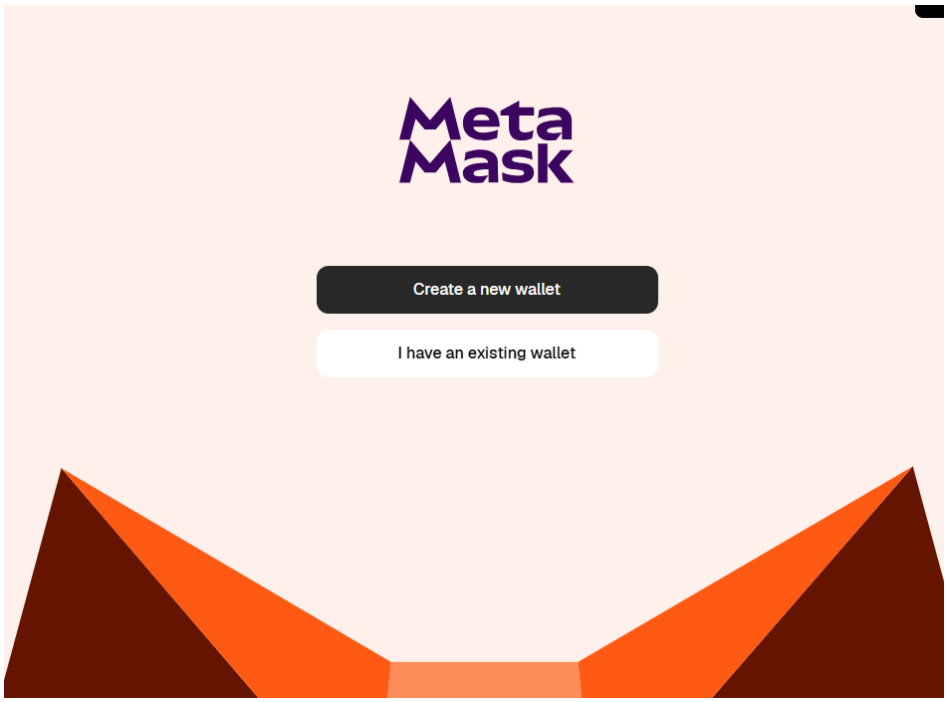
1. Install MetaMask :

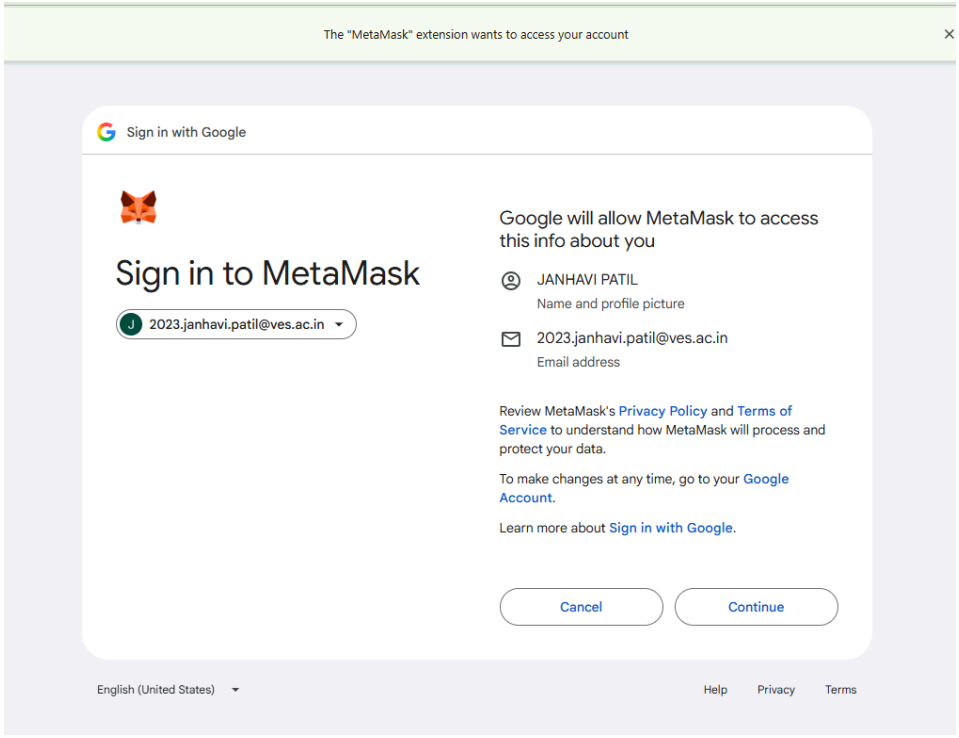
- Follow the Link to install the Google Chrome Extension.





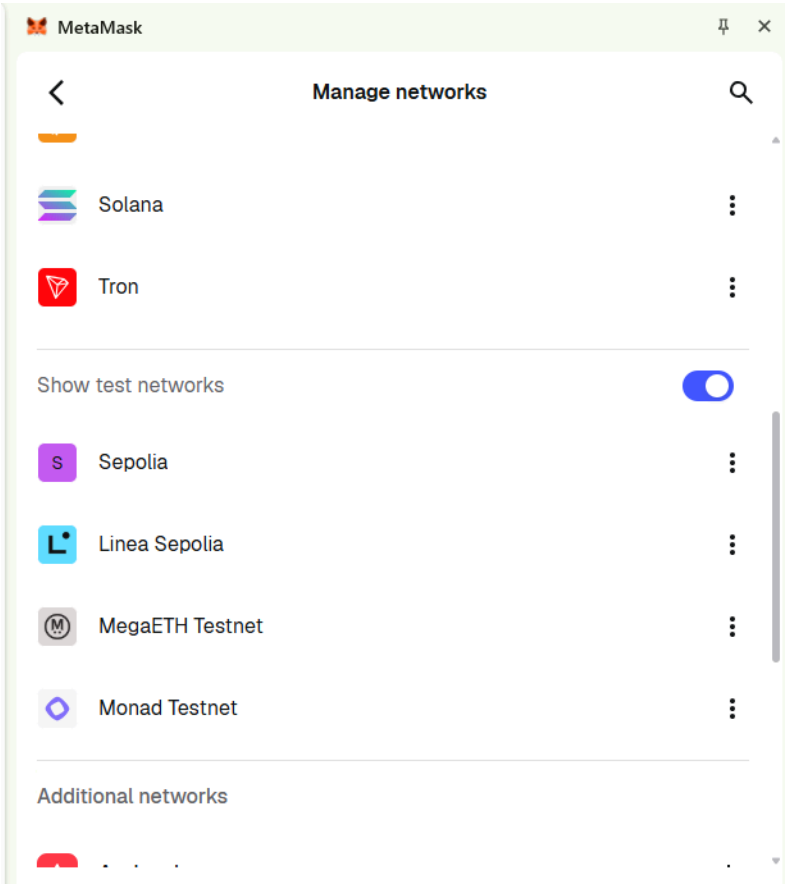
2. Create or Import a Metamask Wallet



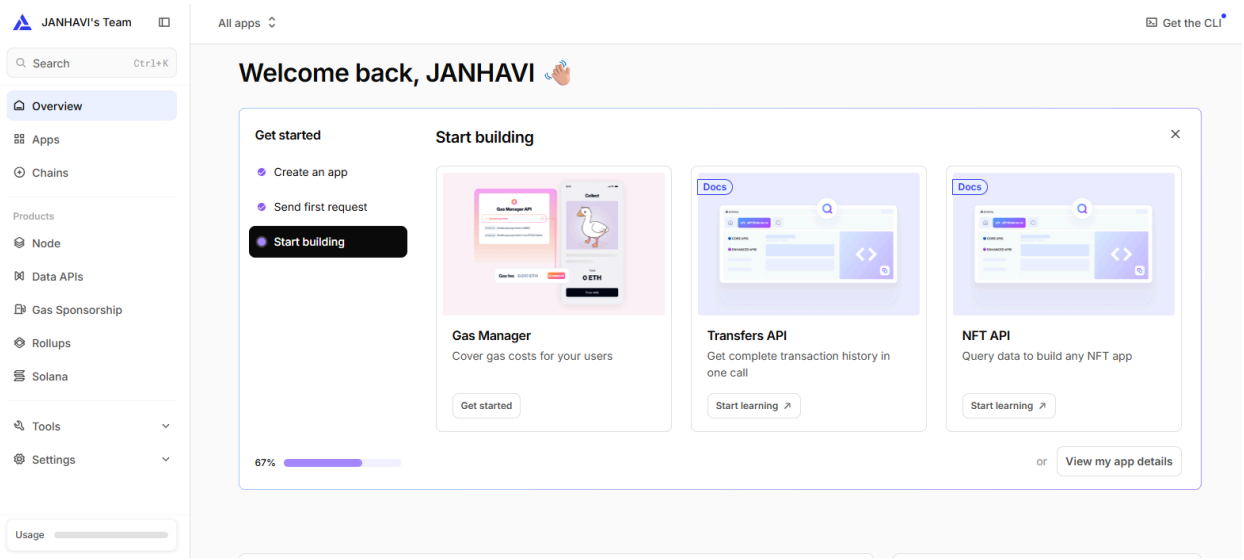


3. Fund Your Wallet from any Testnet:

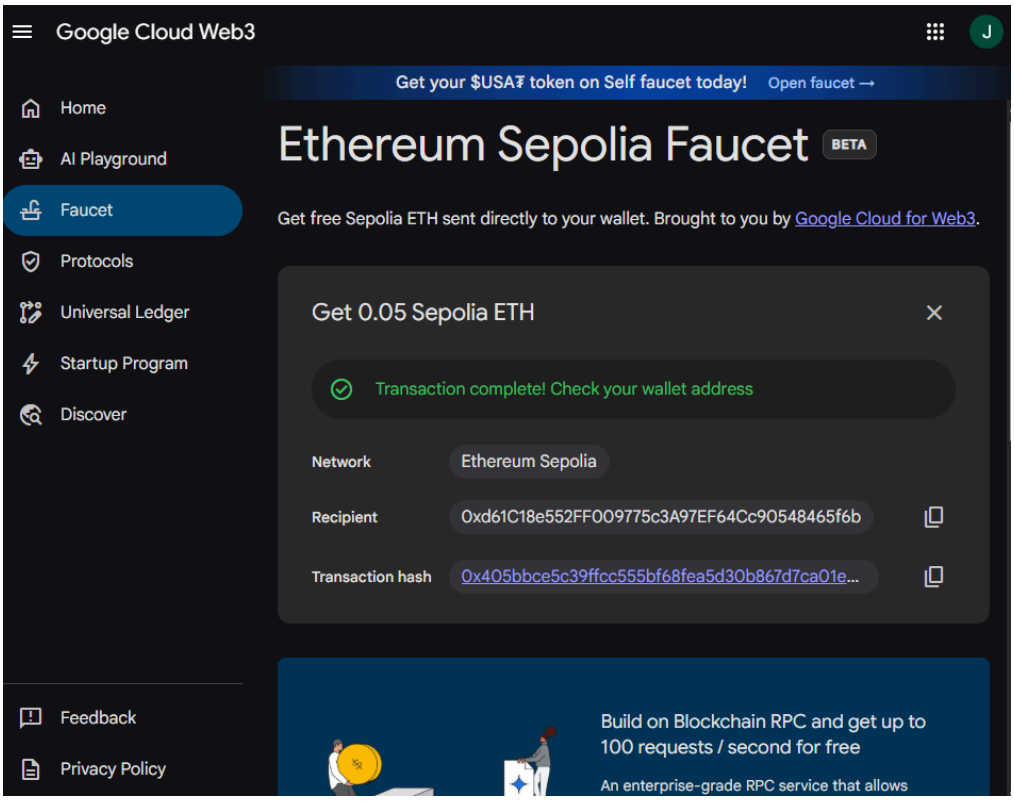
1. Select Sepolia Testnet in Metamask @ https://chrome-extension://nkbihfbeogaeaoehlefnkodbefgpgknn/home.html#



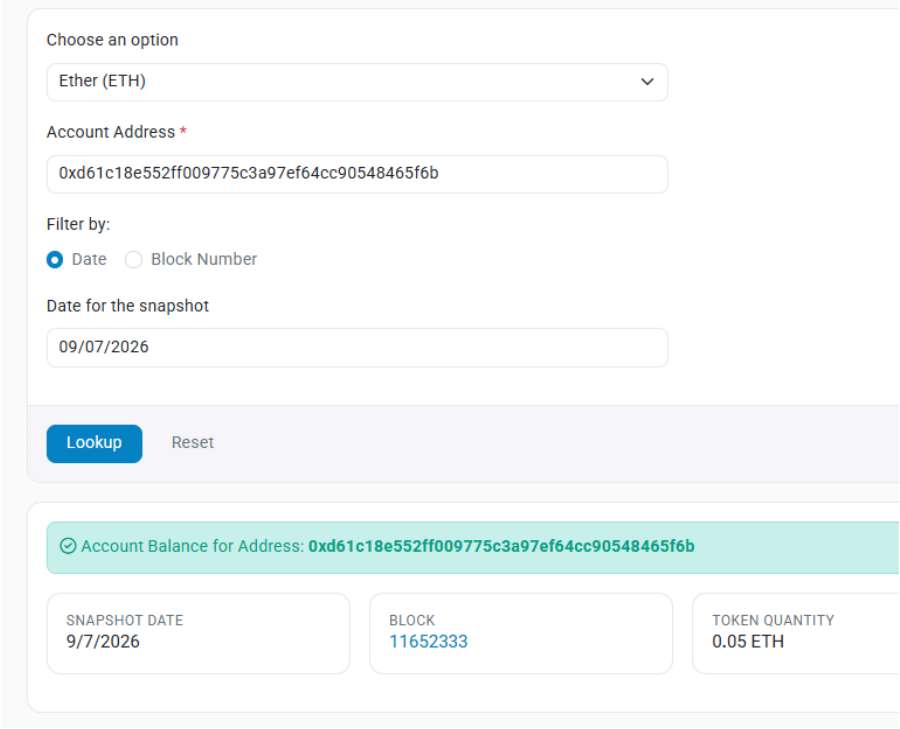
2. Sepolia Testnet details are available on Metamask
3. Create an Alchemy free account @ https://sepoliafaucet.com/



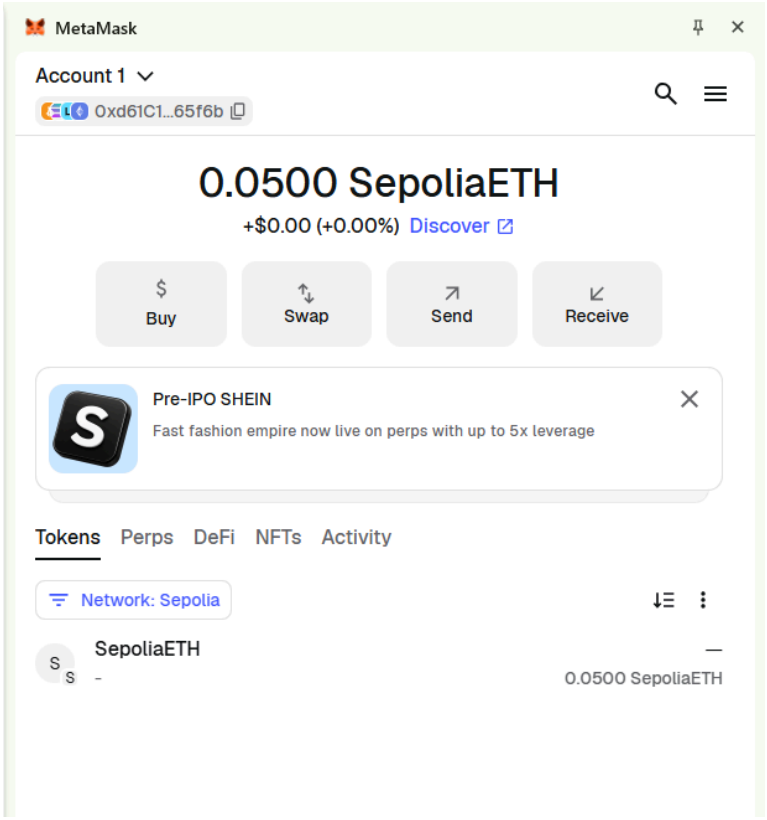
4. Enter your Account details from the Metamask to get free Ethers (only 0.5 ETH per day)



5. After the successful transfer of funds, check the transaction details @ Ehterscan.io https://sepolia.etherscan.io/tx/0x198f961b749df779e322d9e8ebcac66499ab4a457a2dbe5ac6f63202e4a1b840

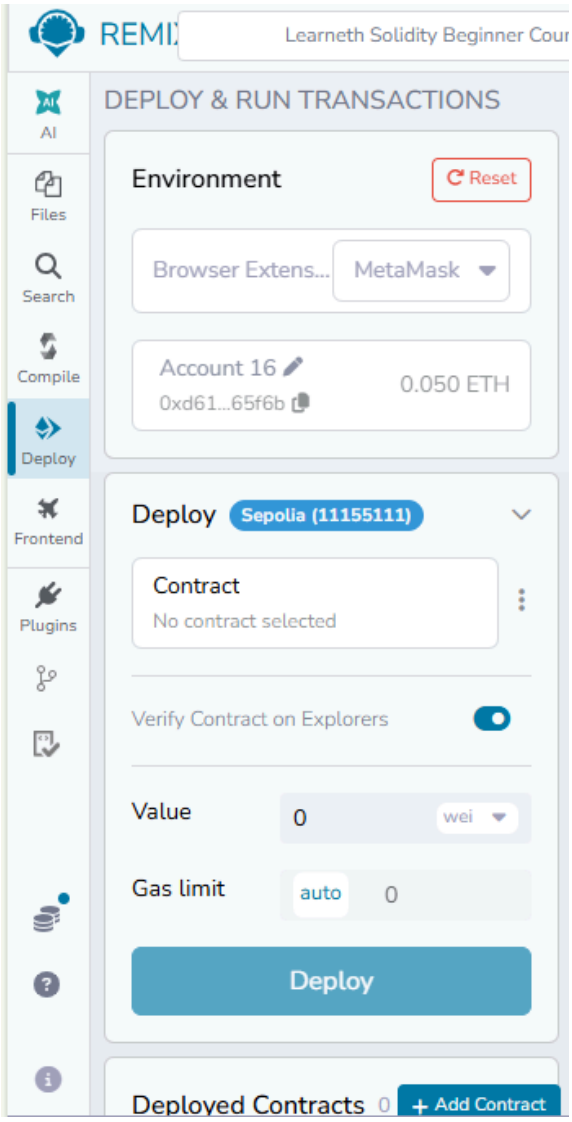


6. The funds are available @ Metamask

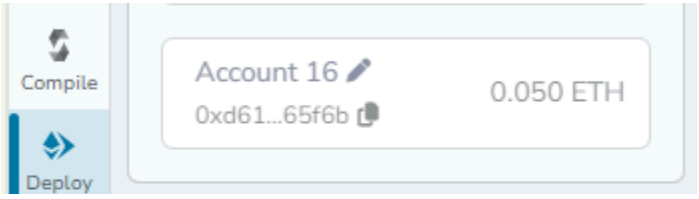


Step - 2 : Connect the Sepolia Testnet / RSK Testnet to Remix IDE

1. In the Remix IDE, select the Environment as Injected Web3.



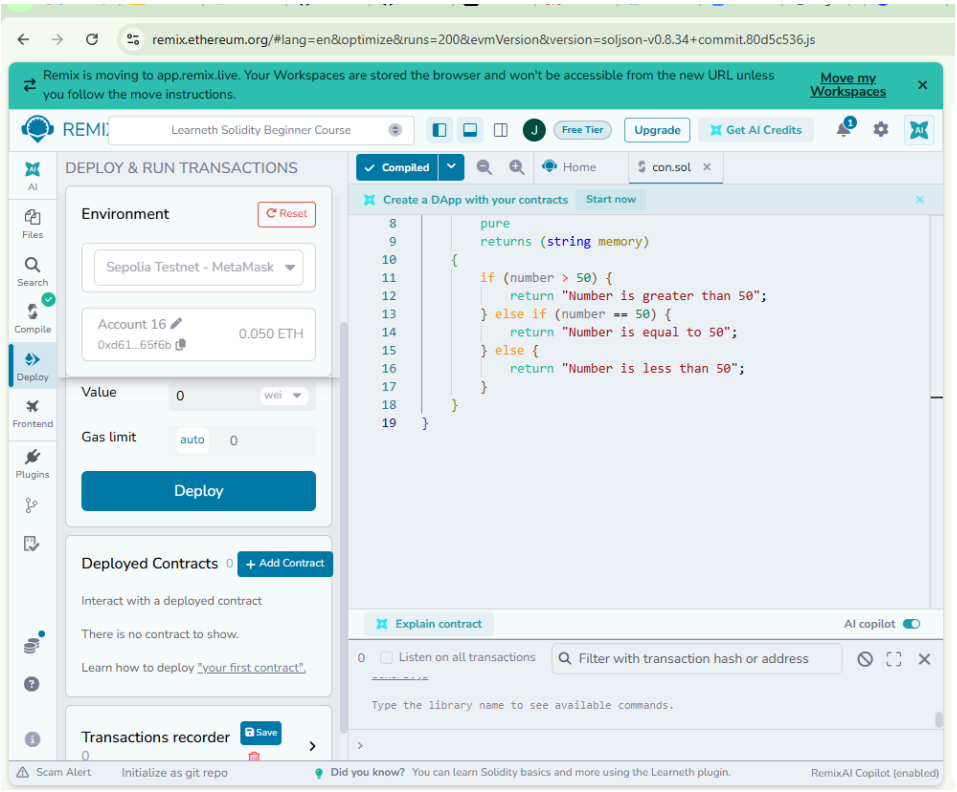
2.Select the account to be linked with and Click Next Button.



3. Click on Connect button. Once the Account is linked to Remix IDE, the Metamask Account details appears in the Deployment Environment as follows:

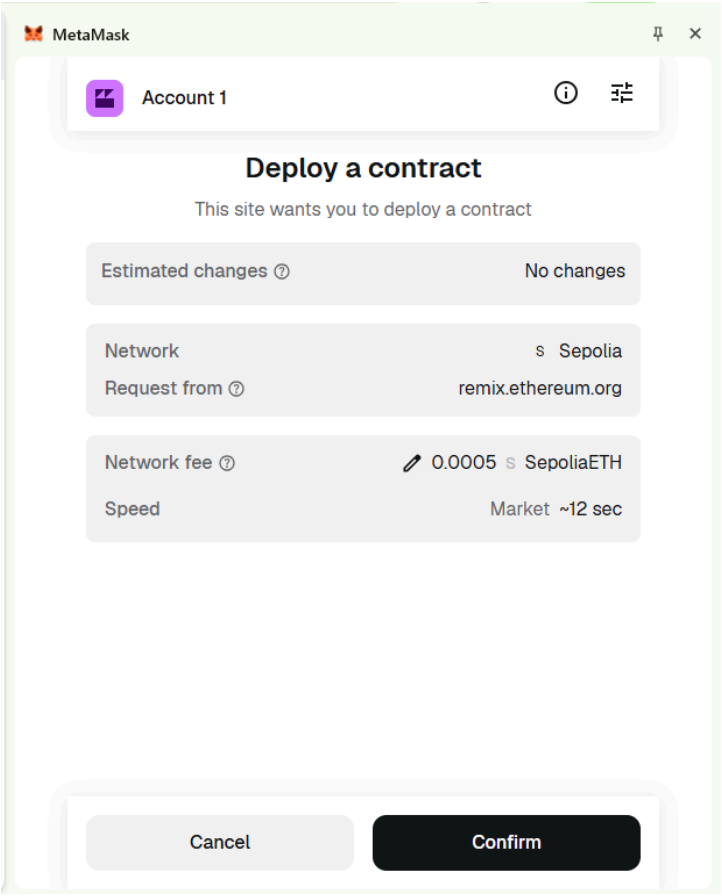


Step - 3 : Create a Simple Solidity Smart Contract

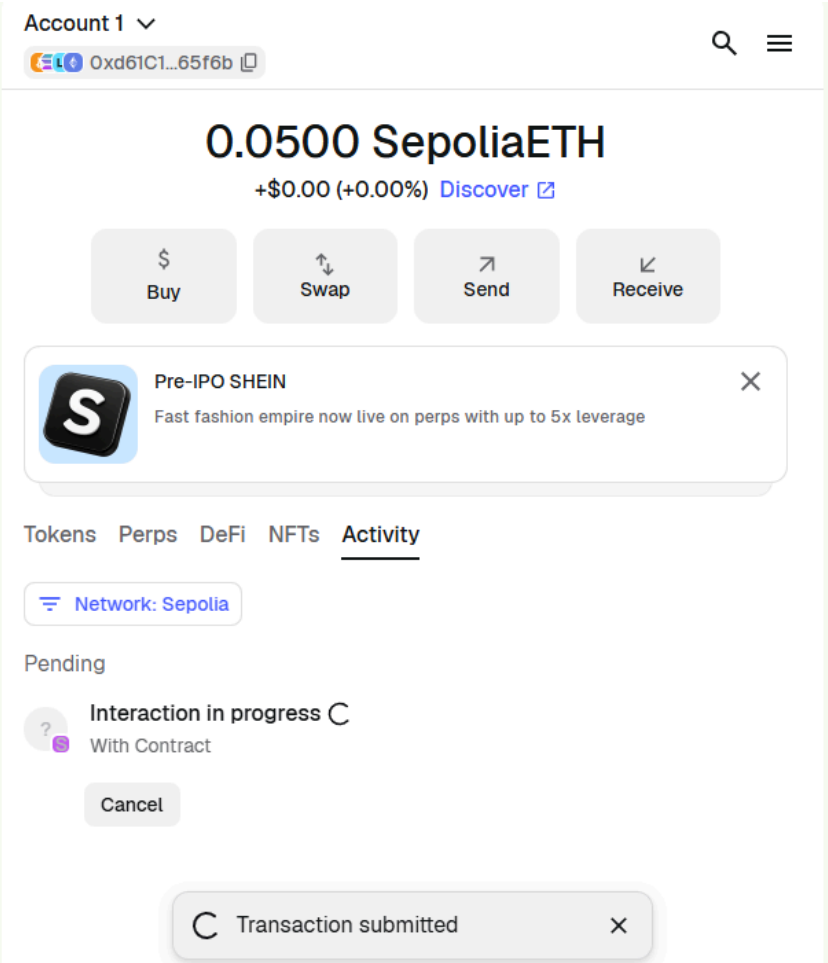


Step - 4 : Compile and Deploy the Smart Contract.

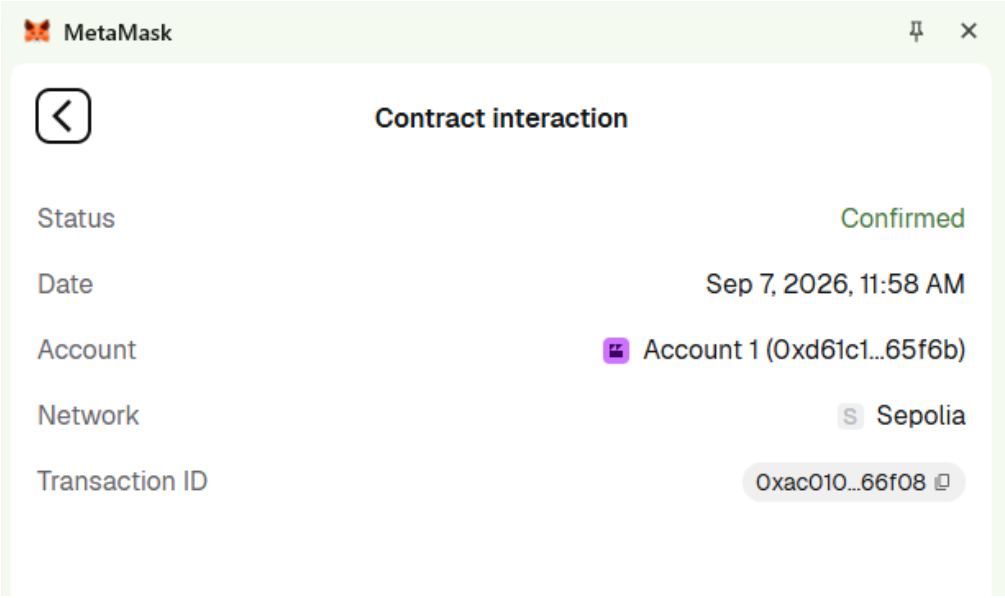
1. Once the Contract is deployed on Remix IDE, the popup appears to confirm the Contract Deployment.



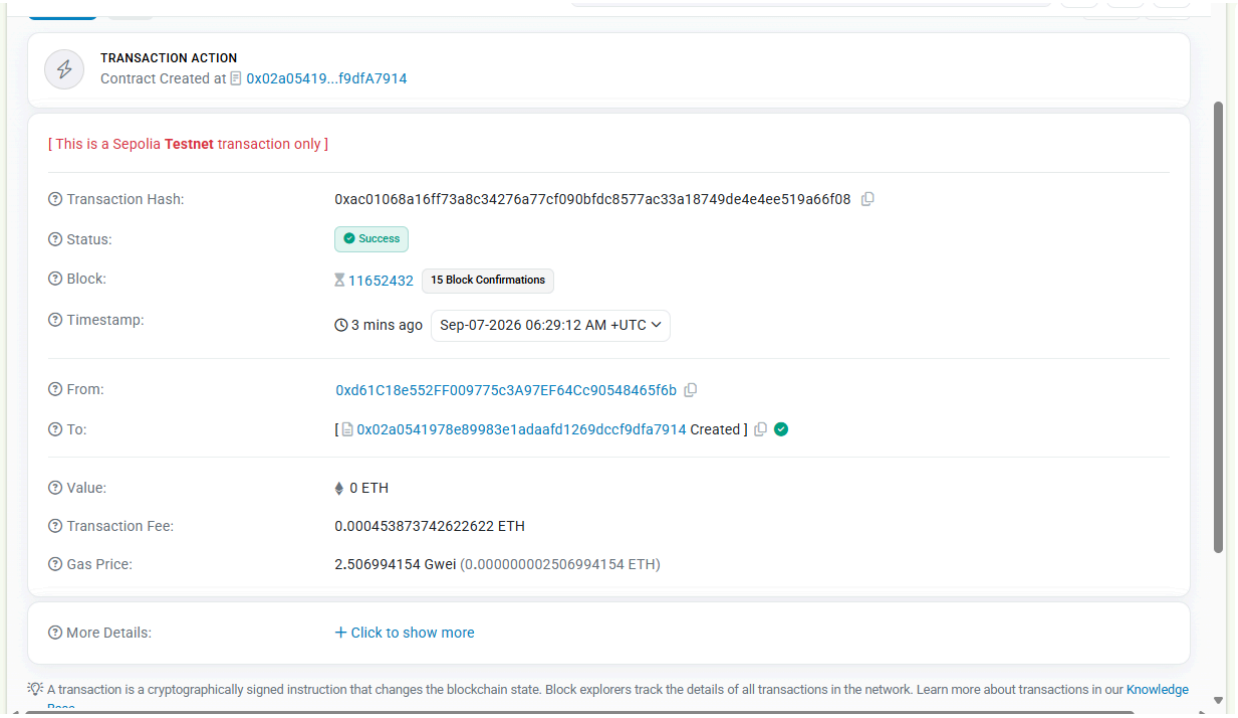
2. As the Deployment is in progress the following popup appears, stating that the deployment is in progress



3. Once the transaction is confirmed the status of the transaction is displayed



4. Transaction Deployment details are displayed, w.r.t the Gas Limit, Used, Gas Used, Gas Price and Total Amount



CONCLUSION:

The smart contract was successfully created, compiled, and deployed using **Solidity and Remix IDE**. MetaMask was successfully integrated with Remix and used to authorize blockchain transactions on the testnet. The experiment provided practical understanding of **smart contracts, MetaMask, testnets, transaction execution, and blockchain interaction**.